2.1 Data Acquisition

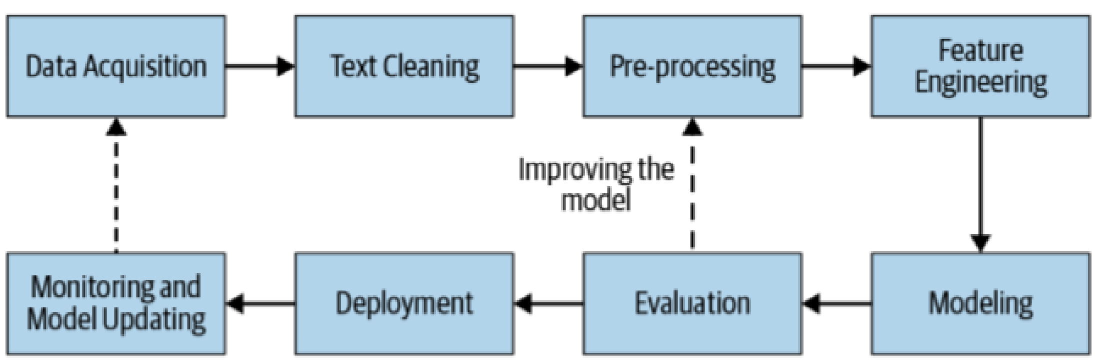
Where the text comes from, and what counts as good enough data.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. The pipeline this week sits in

An applied NLP system is a loop, not a single model.

Stage	Job
Data acquisition	Get text that matches the task
Text cleaning	Pull the words out of HTML, PDF, or other junk
Pre-processing	Put the text into a form you can count or embed
Feature engineering	Turn that text into numbers
Modeling	Fit something that maps numbers to a decision
Evaluation	Measure whether the decision is good enough
Deployment	Plug the module into a larger product
Monitoring	Watch it, then collect more data and start again



Note **2.1** is the first box. Notes **2.2–2.4** are cleaning and pre-processing. Week 3 is features, models, metrics, and production.

The main text is still *Practical Natural Language Processing* (Vajjala, Majumder, Gupta, Surana), Chapter 2.

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2. Data is usually the bottleneck

You can change the model in an afternoon. You cannot invent a labeled corpus in an afternoon.

Example. A chat system must route each incoming message to sales or to customer care. The model is ordinary text classification. The hard part is having enough messages, with the right labels, that look like the messages you will see next month.

In the ideal case you already have thousands or millions of labeled examples from the same product. Then acquisition is done: you split, you train, you evaluate.

Most projects are not that lucky. "Good enough" data means three things at once:

1. **Enough volume** to fit the model you want (a regex needs almost none; a transformer wants a lot).
2. **The right domain.** Product names, slang, and user habits have to match production. A public forum scrape will not contain your SKU codes.
3. **Labels you can trust**, or a plan to get them.

If any of those is missing, you are still in data acquisition.

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3. Where the text comes from

Source	What you get	Watch for
Files you already have	Logs, tickets, PDFs, CSVs, emails	Encoding, PII, messy fields
Public datasets	A starting corpus (Dataset Search)	Domain shift; license
Web pages	HTML you parse yourself	Boilerplate, ToS, drift
APIs	Tweets, news, reviews, transcripts	Rate limits, schema changes, terms
Licensed / purchased data	Cleaner, sometimes labeled	Cost; you still must check fit
Product intervention	Data the product itself collects	Best match to production; needs a live feature

Public data is the first stop. If a dataset is close to the task, build a baseline and measure the gap. Close is not the same as good enough.

Scraping plus human labels is the next stop. It is slow, and the text often lacks the product-specific language you need in production.

APIs are scrape-with-a-contract. You still own cleanup and labeling.

Licensed data (news wires, medical notes, speech transcripts) can save months. Read the license before you train. Some corpora cannot be redistributed or used commercially.

Product intervention is how Google, Facebook, and Netflix grow their own data: ship a feature, log the interaction, label what you can. In industry this is usually the best long-term source, because the distribution *is* production.

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4. When you have too little: augmentation

Start from a small clean set. Make more examples that keep the label.

Trick	What you do
Synonym replacement	Swap a word for a near synonym
Back translation	English → another language → English. Keep the new sentence if it differs
Bigram flip	`going to` → `to going` (noisy; use sparingly)
Entity swap	`I live in California` → `I live in London`
Spelling noise	Replace a word with a near-miss spelling

Advanced options you will hear about: active learning (label the examples the model is unsure of), easy data augmentation, and weak supervision (Snorkel).

Augmentation is not a substitute for in-domain text. It stretches what you have. It does not invent a new domain.

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5. A working rule

Collect the smallest set that is *from the product* and *labeled well*. Use public data and augmentation only to get a baseline or to fill obvious holes. Then go back to the pipeline figure: more data after monitoring is still data acquisition.

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6. Practice

1. For the sales-vs-care router, which is better as training data: 50,000 labeled Wikipedia talk-page comments, or 2,000 labeled chats from the same product? Why?
2. Name one risk of scraping a public Q&A site and treating the posts as if they were your customers.
3. Write one back-translation pair (source sentence and a plausible English rewrite) that would *hurt* a sentiment classifier, not help it.